torch.nn.functional.smooth_l1_loss#

https://pytorch.org/docs/stable/generated/torch.nn.functional.smooth_l1_loss

Description:

Compute the Smooth L1 loss. Function uses a squared term if the absolute element-wise error falls below beta and an L1 term otherwise. See SmoothL1Loss for details. input (Tensor) – Predicted values. target (Tensor) – Ground truth values.

Function Signature

```
torch.nn.functional.smooth_l1_loss(input, target,
size_average=None, reduce=None, reduction='mean', beta=1.0)
[source]#

Parameters

Returns

Return type
```

Returns:

Tensor

Detailed Documentation

Documentation

Compute the Smooth L1 loss.

Function uses a squared term if the absolute element-wise error falls below beta and an L1 term otherwise.

See SmoothL1Loss for details.

input (Tensor) – Predicted values.

target (Tensor) – Ground truth values.

size_average (bool, optional) – Deprecated (see reduction).

reduce (bool, optional) – Deprecated (see reduction).

reduction (str, optional) – Specifies the reduction to apply to the output: ‘none’ | ‘mean’ | ‘sum’. ‘mean’: the mean of the output is taken. ‘sum’: the output will be summed. ‘none’: no reduction will be applied. Default: ‘mean’.

beta (float, optional) – Specifies the threshold at which to change from the squared term to the L1 term in the loss calculation. This value must be positive. Default: 1.0.

L1 loss (optionally weighted).